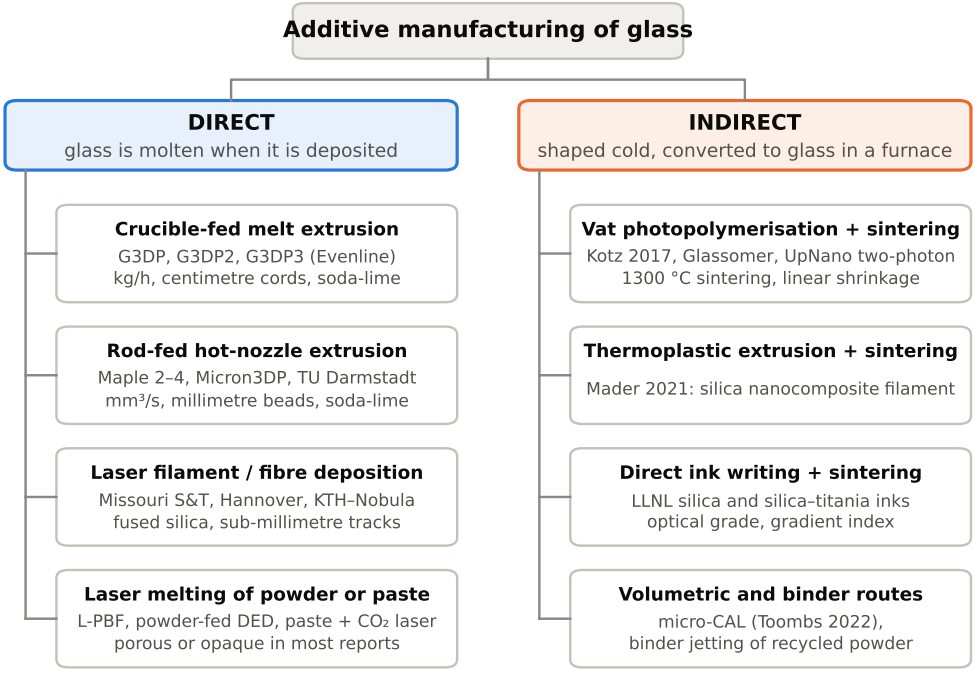
Additive manufacturing of glass
DIRECT
glass is molten when it is deposited
Crucible-fed melt extrusion
G3DP, G3DP2, G3DP3 (Evenline)
kg/h, centimetre cords, soda-lime
Rod-fed hot-nozzle extrusion
Maple 2–4, Micron3DP, TU Darmstadt
mm³/s, millimetre beads, soda-lime
Laser filament / fibre deposition
Missouri S&T, Hannover, KTH–Nobula
fused silica, sub-millimetre tracks
Laser melting of powder or paste
L-PBF, powder-fed DED, paste + CO₂ laser
porous or opaque in most reports
INDIRECT
shaped cold, converted to glass in a furnace
Vat photopolymerisation + sintering
Kotz 2017, Glassomer, UpNano two-photon
1300 °C sintering, linear shrinkage
Thermoplastic extrusion + sintering
Mader 2021: silica nanocomposite filament
Direct ink writing + sintering
LLNL silica and silica–titania inks
optical grade, gradient index
Volumetric and binder routes
micro-CAL (Toombs 2022),
binder jetting of recycled powder